Specification

Customer Name: PLAY-ZONE GmbH

Type : LNM1616BEG2B-1

Customer confirmation		Guangzhou Linong Lighting technology Co.,LTD		
Approval	Confirm	Confirm	Made	Datelanded
			Lingshu	2013-12-18

- **Description of the type：**

LNM XXXX XX XX

① ② ③ ④

1 Stand for the dot matrix series of Guangzhou Linong Lighting Technology Co., LTD
2 Stand for the code of every small point' s size and the quantity of line and row
3 Stand for the polarity and emitting colour
4 Stand for the length of the PIN and the colour of encapsulation adhesive

- **Description：**

1 Description: Φ3 mm 16*16 the Red and Green colour Common Anode dot matrix
2 Type: LNM1616BEG2B-1
3 Overall dimension: 64*64*7.5 （mm）
4 Emitting colour: Red and Green
5 Encapsulation adhesive colour: tansparent adhesive
6 Surface colour of the shell: Black
7 Diameter of the Pin: Φ0.6±0.05mm

- **Certification ：**

1 ISO9001 ： 2008 ；
2 SGS

- **Technical terms ：**

1 Wavelength: The data which use to describe the changes of the colour of the light can be call wavelength.The range of the wavelength between 380 to 780 nm can be see by human eye . Unit: Nm.
2 Luminance: Luminance is a photometric measure of the luminous intensity per unit area of light travelling in a given direction. It describes the amount of light that passes through or is emitted from a particular area,and falls within a given solid angle. Unit: Nit.
3 Luminous intensity: Refers to the degree of the bright light. It is a physical quantity also indicates the visible radiation of the strength of the illuminant. Unit:Cd.
4 Luminous flux: Luminous flux is the measure of the perceived power of light. It differs from radiant flux,the measure of total power of light emitted,in that luminous flux is adjusted to reflect the varying sensitivity of the human eye to different wavelengths of light. Unit: Lm.

- **ELECTRICAL/OPTICAL （ chip ） parameter：**

Red

Ta=25℃

parameter	Symbol	Test condition	MIN.	TYP.	MAX.	UNIT
Forward Voltage	Vf	10mA	_____	2.05	____	V
Reverse Current	Ir	VF=5V	_____	____	10	uA
Luminous Intensity	Iv	20mA	8.5	----	14	mcd
Wavelength	λ	10mA	620	----	629	nm

Absolute rating

Ta=25℃

parameter	Symbol	MIN.	MAX.	UNIT
Forward Current	If	_____	30	mA
Peak Forward Current	Ifp	_____	80	mA
Reverse Voltage	Vr	_____	5	V
Operating Temperature	Topr	-30	80	℃
Storage Temperature	Tstg	-40	85	℃
Power Dissipation	Pd	_____	75	mW

Green

Ta=25℃

parameter	Symbol	Test condition	MIN.	TYP.	MAX.	UNIT
Forward Voltage	Vf	10mA	___	2.1	___	V
Reverse Current	Ir	VF=5V	___	___	10	uA
Luminous Intensity	Iv	20mA	14.4	---	18	mcd
Wavelength	λ	10mA	571	---	572	nm

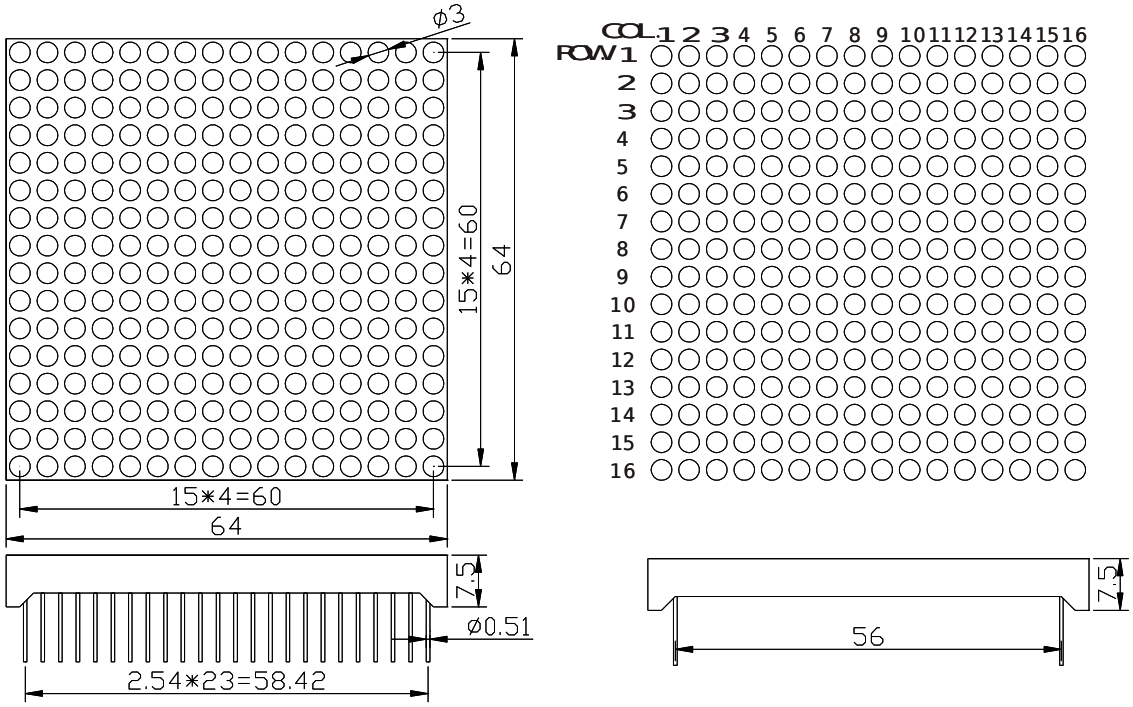
Absolute rating

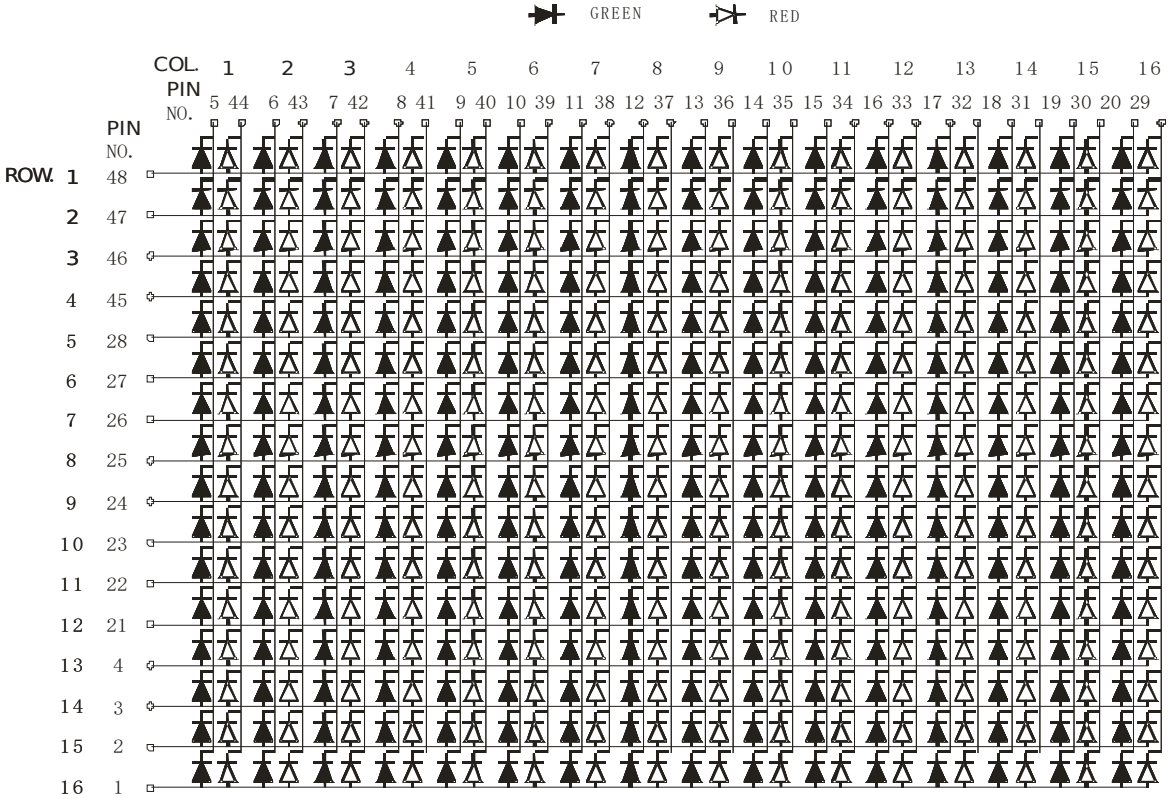
Ta=25℃

parameter	Symbol	MIN.	MAX.	UNIT
Forward Current	If	_____	30	mA
Peak Forward Current	Ifp	_____	80	mA
Reverse Voltage	Vr	_____	5	V
Operating Temperature	Topr	-30	80	℃
Storage Temperature	Tstg	-40	85	℃
Power Dissipation	Pd	_____	75	mW

***Remark: pulse width:10ms Duty Cycle :1/10**

- **Overall dimension and the function of certain PIN：**





remark：

1 **All dimensionsUnit are mm;**

2 **Tolerance are ±0.25mm unless otherwise noted.**

- **Storerage and transportation：**

1 Operating Temperature -30 ～80℃

2 Storage Temperature -40 ～85℃

3 Need moistureproof

4 When transporting , try to reduce the chance of mechanical vibration and shock:; when carrying , handle gently, Do not stacking too high, Do not put overweight goods on the surface of the LED.

- **Welding：**

Welding condition：

1 Soldering iron welding: Soldering iron （MAX.30W）, tip temperature should not more than 260℃; welding time should not more than 3 Seconds

2 Wave-soldering: Dip soldering maximum temperature is 260℃; dip soldering should not more than 3 Seconds.

Soldering iron welding	1.Maximum soldering iron power: 30W
	2.Maximum temperature: 260℃
	3.Maximum duration: 3seconds
Wave -soldering	1.Maximum temperature of perheating: 100℃
	2.Maximum preheating duration: 60seconds
	3.Dip soldering maximum temperature: 260℃
	4.Dip soldering duration: 3seconds

- **Clean：**

Some chemicals will do harm to the surface of the LED such as : triclene,acetone. Can use ethanol to wipe (not more than 3 minutes)under the normal temperature.

- **Warning**

1 Adopt the right current and voltage when operating.

2 The product can not storage and use under the corrosion gas and can not exposure in the air for a long time, otherwish, it will get oxidation.

3 All the equipments and instruments connected with the LED should be grounded.

4 All the people who touch the LED should wear ESD wrist strap or antistatic gloves.

5 Once the LED damaged by the static it will show some bad characteristics such as :Leakage current is add, static consequent voltage rise, does not give out normal light or not shine under the low current, the service life will be shorten.